

Medvale Internal Medicine

Chapter 1I: Pericardial and Myocardial Disease

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Why this matters clinically

Pericardial and myocardial diseases sit in the overlap between chest pain, dyspnea, arrhythmia, shock, and heart failure. A patient with pericarditis may look like a patient with acute coronary syndrome because both can have chest pain, ST-segment abnormalities, and elevated inflammatory markers. A patient with myocarditis may look like a viral syndrome, an MI mimic, acute heart failure, or a malignant arrhythmia syndrome. A patient with cardiac tamponade may initially look like nonspecific hypotension until venous congestion, pulsus paradoxus, and echocardiographic physiology reveal obstructive shock. For internal medicine reasoning, these diseases are not isolated memorization topics; they are nodes that connect chest pain, ECG interpretation, cardiomyopathy, heart failure, sudden death, and shock physiology.

The uploaded cardiology section emphasizes hypertrophic cardiomyopathy, restrictive cardiomyopathy, myocarditis, acute pericarditis, constrictive pericarditis, pericardial effusion, and cardiac tamponade, including classic numbers such as acute pericarditis diagnostic criteria, effusion detection thresholds, pulsus paradoxus greater than 10 mm Hg, and rapid 200 mL tamponade physiology. This chapter reorganizes those signals into a clinical reasoning framework. The goal is to teach how to recognize the pattern, identify the danger physiology, and choose management based on hemodynamic risk.

Graph-RAG anchor

Pericardial disease is best indexed around the concepts **inflammation, fluid accumulation, restricted filling, and tamponade physiology**. Myocardial disease is best indexed around **myocyte inflammation, hypertrophy, dilation, restriction, arrhythmia risk, and heart failure phenotype**.

Numbers, Cutoffs, and Timelines

Topic	High-yield number or cutoff	Clinical meaning
HCM inheritance	Often autosomal dominant	Family history and sudden death screening matter.
HCM obstruction gradient	≥ 30 mm Hg suggests LVOT obstruction; ≥ 50 mm Hg often used as a treatment threshold in symptomatic patients	Gradient helps classify obstructive physiology and guide septal reduction decisions.
Acute pericarditis diagnosis	At least 2 of 4 criteria	Chest pain, rub, ECG changes, or effusion.
Acute pericarditis recovery	Often 1-3 weeks; uncomplicated cases may resolve in 2-6 weeks	Persistent or recurrent symptoms require reassessment.
Tamponade complication in acute pericarditis	Source notes up to about 15%	Close observation is needed when effusion is present or symptoms worsen.
Echo effusion sensitivity	Can detect as little as about 20 mL of fluid	Echo is preferred for effusion and tamponade physiology.
CXR silhouette enlargement	Usually requires >250 mL of pericardial fluid	Chest x-ray is insensitive for small effusions.

Small insignificant effusion follow-up	Repeat echo in about 1-2 weeks	Useful when stable but etiology or size progression is uncertain.
Rapid tamponade volume	As little as about 200 mL if accumulation is rapid	Rate of accumulation matters more than absolute volume.
Slow effusion volume	May reach about 2 L before tamponade if gradual	The pericardium stretches over time.
Pulsus paradoxus	>10 mm Hg inspiratory systolic BP drop	Classic bedside clue to tamponade physiology.
Constrictive pericarditis AF frequency	Less than half in source framing	AF may occur but is not required.
HCM exercise advice	Avoid intense competitive athletics in many symptomatic or high-risk patients	Dehydration and increased contractility can worsen obstruction.

Core Illness Scripts

An illness script is a compressed clinical pattern: mechanism, patient context, presentation, diagnostic clue, and management direction. For this section, the highest-yield scripts are hypertrophic cardiomyopathy, restrictive cardiomyopathy, myocarditis, acute pericarditis, constrictive pericarditis, pericardial effusion, and tamponade. These conditions overlap but differ by the central physiologic problem. HCM produces a stiff and sometimes obstructed ventricle. Restrictive cardiomyopathy produces impaired filling from infiltrative or fibrotic myocardium. Myocarditis produces inflammatory myocyte injury. Pericarditis produces inflamed pericardial layers that hurt with motion and respiration. Constrictive pericarditis produces a rigid external shell around the heart. Effusion produces fluid around the heart. Tamponade occurs when that fluid causes clinically important impaired diastolic filling.

Disease	Core mechanism	Classic clinical pattern
Hypertrophic cardiomyopathy	Hypertrophied, stiff LV; sometimes dynamic LV outflow tract obstruction	Exertional dyspnea, chest pain, syncope, systolic murmur that increases with Valsalva/standing.
Restrictive cardiomyopathy	Noncompliant infiltrated or fibrotic myocardium limits filling	Right-sided and left-sided HF signs with preserved or variable systolic function.
Myocarditis	Inflammation and injury of cardiac myocytes	Viral prodrome plus chest pain, troponin elevation, arrhythmia, or HF.
Acute pericarditis	Inflamed pericardial surfaces rub and generate pleuritic positional pain	Sharp chest pain relieved by leaning forward, friction rub, diffuse ST elevation/PR depression.
Constrictive pericarditis	Rigid pericardium abruptly limits diastolic filling	JVD, Kussmaul sign, edema/ascites, pericardial knock, equalized diastolic pressures.
Pericardial effusion	Fluid collects in pericardial space	Often silent; may cause muffled sounds, enlarged cardiac silhouette, or tamponade if pressure rises.
Cardiac tamponade	Pericardial pressure impairs chamber filling	Hypotension, JVD, pulsus paradoxus, narrow pulse pressure, obstructive shock.

Hypertrophic Cardiomyopathy

Hypertrophic cardiomyopathy is a disease of excessive myocardial hypertrophy that cannot be fully explained by loading conditions such as hypertension or aortic stenosis. The classic mechanism is sarcomere dysfunction, often inherited in an autosomal dominant pattern, although spontaneous mutations occur. The ventricle becomes thick and stiff, so diastolic filling becomes impaired even when systolic contraction appears vigorous. In obstructive HCM, asymmetric septal hypertrophy and systolic anterior motion of the mitral valve narrow the left ventricular outflow tract. This obstruction is dynamic, which means it changes with preload, afterload, heart rate, and contractility.

The dynamic nature of the obstruction explains the physical exam. Maneuvers that reduce left ventricular volume, such as standing or the Valsalva maneuver, make the LV cavity smaller and worsen obstruction. The murmur becomes louder. Maneuvers that increase left ventricular volume or systemic vascular resistance, such as squatting or sustained handgrip, tend to reduce the gradient and soften the murmur. This is the opposite of many fixed outflow lesions. The same physiology explains why dehydration, vigorous exertion, tachycardia, and positive inotropic drugs can worsen symptoms.

Symptoms include exertional dyspnea, exertional chest pain, syncope or presyncope, palpitations, atrial fibrillation, ventricular arrhythmias, and sudden cardiac death. Some patients are asymptomatic for many years, while others first present after a syncopal episode or family screening. A sustained point of maximal impulse, loud S4, bifid carotid upstroke, and a systolic ejection murmur at the left lower sternal border can support the diagnosis. Echocardiography establishes the diagnosis, defines septal thickness and mitral valve motion, and estimates the LVOT gradient. Contemporary HCM guidance uses a peak gradient of at least 30 mm Hg to identify obstruction, while resting or provoked gradients of at least 50 mm Hg often guide advanced therapy for refractory symptoms.

Treatment logic: HCM

The treatment goal is not to increase contractility; the goal is to improve filling and reduce obstruction. Non-vasodilating beta blockers are typical first-line therapy for symptomatic obstructive HCM. Verapamil or diltiazem can be considered if beta blockers are not tolerated, but vasodilators and aggressive diuresis can worsen obstruction by reducing preload or afterload. Patients with high-risk features may need ICD evaluation. Septal myectomy or alcohol septal ablation can be considered when symptoms remain severe despite medical therapy and obstruction is significant.

Restrictive Cardiomyopathy

Restrictive cardiomyopathy is a myocardial filling disorder. The ventricle is not necessarily dilated, and systolic function may be preserved until late disease, but the myocardium is stiff and noncompliant. During diastole, ventricular pressure rises quickly after only a small amount of filling. This produces pulmonary congestion, systemic venous congestion, exercise intolerance, atrial enlargement, and arrhythmias. Because the restriction is inside the myocardium, restrictive cardiomyopathy can resemble constrictive pericarditis, where the restriction is outside the heart. The distinction matters because constrictive pericarditis may be surgically treatable, whereas restrictive cardiomyopathy is treated by addressing the underlying myocardial disease and managing heart failure symptoms carefully.

Important causes include amyloidosis, sarcoidosis, hemochromatosis, scleroderma, carcinoid syndrome, radiation injury, chemotherapy-related injury, and idiopathic disease. Amyloidosis can produce a thickened ventricular wall on echocardiography with low-voltage ECG, a discordance that is highly suggestive because the apparent wall thickening reflects infiltrative protein rather than normal hypertrophy. Hemochromatosis can cause cardiomyopathy through iron deposition and should connect in a graph database to liver disease, diabetes, bronze skin, arthropathy, and elevated iron indices. Sarcoidosis connects cardiomyopathy to conduction disease, ventricular arrhythmias, granulomatous lung disease, and sudden death.

Treatment depends on cause. Hemochromatosis may require phlebotomy or iron chelation. Sarcoidosis may require immunosuppression for active inflammation. Amyloidosis requires subtype-specific therapy and may require organ-specific management. Diuretics can reduce congestion but must be used carefully because excessive preload reduction can lower cardiac output in a stiff ventricle. This caution parallels constriction and tamponade: when filling is constrained, the patient can become preload sensitive.

Myocarditis

Myocarditis is inflammation of the myocardium. It may be caused by viruses such as Coxsackie virus, parvovirus B19, and human herpesvirus 6; bacteria such as group A Streptococcus, Lyme disease, and Mycoplasma; autoimmune disease such as systemic lupus erythematosus; toxins; medications; or idiopathic inflammatory injury. The classic teaching pattern is a young patient with a recent viral syndrome who develops chest pain, fatigue, dyspnea, palpitations,

syncope, heart failure, or arrhythmia. However, myocarditis can occur across age groups and may range from mild self-limited disease to fulminant cardiogenic shock.

Myocarditis is clinically important because it can mimic myocardial infarction. Patients may have chest pain, ST-T changes, and elevated troponin because inflamed myocytes leak cardiac biomarkers. The difference is that the injury is inflammatory rather than primarily due to acute coronary plaque rupture. Myocarditis can also mimic dilated cardiomyopathy when inflammation damages contractile function. In a graph-RAG structure, myocarditis should connect to viral prodrome, troponin elevation, arrhythmias, acute heart failure, cardiac MRI, and supportive care.

The initial workup often includes ECG, troponin, inflammatory markers, echocardiography, and cardiac MRI when available and appropriate. Endomyocardial biopsy is not required for every suspected case, but modern expert consensus emphasizes that biopsy can confirm specific forms such as giant cell, eosinophilic, or sarcoid myocarditis and can identify some infectious agents. Treatment is mainly supportive, with heart failure therapy, arrhythmia management, avoidance of intense exercise during active disease, and treatment of an identified cause when possible. Fulminant myocarditis may require ICU care and mechanical circulatory support.

Acute Pericarditis

Acute pericarditis is inflammation of the pericardial sac. It may be idiopathic or post-viral, and it may also result from infection, autoimmune disease, uremia, malignancy, trauma, radiation, medications, post-MI syndromes, or post-pericardiotomy inflammation. The pain is typically sharp, pleuritic, and positional. It worsens when the patient lies flat, coughs, swallows, or takes a deep breath, and it improves when the patient sits up and leans forward. Pain may radiate to the trapezius ridge or neck because the phrenic nerve innervates the pericardium.

The diagnosis is classically made when at least 2 of 4 criteria are present: typical pleuritic positional chest pain, a pericardial friction rub, ECG changes consistent with pericarditis, or a new/worsening pericardial effusion. A rub is very specific but can be transient. It may have up to three components corresponding to atrial systole, ventricular systole, and early diastole, and it is often best heard with the patient leaning forward at end-expiration. ECG changes can evolve through stages. Early acute pericarditis often shows diffuse ST elevation and PR depression, with reciprocal PR elevation and ST depression in aVR. Unlike STEMI, the ST elevation is usually diffuse rather than confined to one coronary territory.

Most uncomplicated cases are treated with high-dose NSAIDs plus colchicine, with gastroprotection and monitoring when appropriate. Glucocorticoids can relieve pain but are generally avoided when possible because they are associated with recurrent pericarditis and may be inappropriate for infectious causes. Patients with low-risk uncomplicated pericarditis can often be treated as outpatients, but fever, leukocytosis, large effusion, tamponade signs, immunosuppression, trauma, anticoagulation, subacute course, or failure to respond should push toward admission and further evaluation.

Acute pericarditis criterion	Clinical clue
Typical chest pain	Sharp, pleuritic, positional, better sitting up/leaning forward.
Friction rub	Specific, transient, scratchy/high-pitched, best with patient leaning forward.
ECG changes	Diffuse ST elevation and PR depression; PR elevation/ST depression in aVR.
Effusion	Seen on echocardiogram or sometimes suggested by CXR if large.

Constrictive Pericarditis

Constrictive pericarditis occurs when the pericardium becomes fibrotic, thickened, calcified, or rigid enough to limit diastolic filling. In early diastole, blood enters quickly, but filling then abruptly stops when the expanding heart meets the fixed pericardial shell. This produces the classic hemodynamic idea of rapid early filling followed by sudden halt. Clinically, patients can look like they have right-sided heart failure, cirrhosis, or nephrotic syndrome because they develop edema, ascites, hepatic congestion, abdominal fullness, and elevated jugular venous pressure.

The most prominent physical finding is often JVD. Kussmaul sign, an inspiratory rise or failure of the JVP to fall, reflects impaired right-sided filling during inspiration. A pericardial knock can occur shortly after S2 when ventricular filling abruptly stops. Constrictive pericarditis may follow recurrent pericarditis, prior cardiac surgery, radiation therapy, tuberculosis, malignancy, connective tissue disease, uremia, trauma, or may be idiopathic. It can be difficult to distinguish from restrictive cardiomyopathy. Echocardiography, CT, MRI, and cardiac catheterization may be needed. Catheterization can show elevated and equalized diastolic pressures and ventricular pressure tracings with a rapid y descent and dip-and-plateau or square-root pattern.

Treatment begins with addressing the underlying cause and using diuretics to relieve congestion. However, definitive treatment for symptomatic chronic constriction may be surgical pericardiectomy. The decision is high stakes because pericardiectomy can improve hemodynamics but carries significant operative risk, especially in advanced disease or radiation-associated constriction.

Original diagram: constriction versus restriction

Constrictive pericarditis: rigid pericardial shell outside the heart -> equalized diastolic pressures -> JVD, Kussmaul sign, pericardial knock -> potentially surgical pericardiectomy.

Restrictive cardiomyopathy: stiff infiltrated myocardium inside the heart -> impaired filling with myocardial disease -> atrial enlargement, arrhythmias, HF -> treat infiltrative/systemic cause.

Shared node: diastolic filling limitation -> preload sensitivity -> congestion with relatively preserved systolic function early.

Pericardial Effusion

A pericardial effusion is fluid in the pericardial space. It can occur from acute pericarditis, malignancy, uremia, hypothyroidism, trauma, post-procedure bleeding, autoimmune disease, heart failure-related volume overload, cirrhosis, nephrotic syndrome, or idiopathic inflammation. Many effusions are asymptomatic and discovered when imaging is performed for another reason. Others cause chest pressure, dyspnea, cough, dysphagia, hoarseness, or signs of tamponade. The clinical risk depends less on the absolute amount of fluid and more on the rate of accumulation and pericardial compliance.

Echocardiography is the imaging procedure of choice because it confirms the effusion, estimates size, and detects tamponade physiology. The uploaded source notes that echo can show as little as about 20 mL of fluid. Chest x-ray is insensitive for small effusions and usually shows an enlarged cardiac silhouette only when more than about 250 mL has accumulated; a classic water-bottle configuration may appear in large chronic effusions. ECG may show low QRS voltage or electrical alternans, but ECG findings are not sensitive or specific enough to diagnose or exclude effusion.

Pericardiocentesis is not automatically indicated for every effusion. The decision depends on hemodynamic stability, suspected etiology, size, symptoms, and tamponade features. Pericardial fluid analysis can include protein, glucose, cell count and differential, cytology, hematocrit, Gram stain and culture, acid-fast testing, mycobacterial PCR, fungal studies, and LDH depending on clinical suspicion. If an effusion is small and clinically insignificant, repeat echocardiography in 1-2 weeks may be appropriate to confirm stability or resolution.

Cardiac Tamponade

Cardiac tamponade is a clinical syndrome in which pericardial fluid pressure impairs diastolic filling and reduces stroke volume. The key teaching point is that the rate of accumulation matters more than the total volume. A rapid 200 mL hemopericardium from trauma or free wall rupture can cause tamponade because the pericardium has no time to stretch. A slowly accumulating effusion may reach 2 liters before tamponade develops because the pericardial sac gradually adapts.

The physiology is obstructive shock. Pressure rises in the pericardial space and compresses the cardiac chambers, especially the low-pressure right atrium and right ventricle. During diastole, pressures in the right ventricle, left ventricle, right atrium, left atrium, pulmonary artery, and pericardial space can become elevated and equalized. Ventricular filling is impaired, stroke volume falls, pulse pressure narrows, and compensatory tachycardia develops. Beck triad -

hypotension, muffled heart sounds, and elevated JVP - is classic but not always complete. Pulsus paradoxus, defined as an inspiratory systolic blood pressure drop greater than 10 mm Hg, is another high-yield clue.

Tamponade is ultimately a clinical diagnosis supported by echocardiography. Echo is the most sensitive and specific noninvasive test for tamponade physiology and may show chamber collapse, exaggerated respiratory variation in inflow velocities, a plethoric IVC, and pericardial effusion. Electrical alternans can occur when the heart swings within a large effusion, but ECG cannot rule tamponade in or out. If the patient is unstable, urgent pericardiocentesis is indicated unless the cause requires immediate surgery, such as traumatic hemopericardium or post-MI free wall rupture. In hemorrhagic traumatic tamponade, pericardiocentesis may be only a temporizing measure and should not delay definitive surgical control of bleeding.

Original diagram: tamponade pressure sequence

Rapid fluid or blood enters pericardial space
 -> pericardial pressure rises
 -> right-sided chambers collapse first because they are lower pressure
 -> diastolic filling falls
 -> stroke volume falls and pulse pressure narrows
 -> sympathetic compensation causes tachycardia and JVD
 -> obstructive shock if not relieved

Clinical Differentiation: Pericarditis, MI, PE, and Dissection

Pericarditis, MI, pulmonary embolism, and aortic dissection can all present with chest pain, tachycardia, and distress. Pericarditis pain is often sharp, pleuritic, positional, and improved by leaning forward. STEMI pain is usually pressure-like, persistent, associated with diaphoresis or nausea, and localized to a coronary ECG territory. Pulmonary embolism often causes pleuritic pain with dyspnea, tachycardia, hypoxemia, and risk factors for thrombosis. Aortic dissection often causes abrupt tearing pain radiating to the back, pulse or blood pressure asymmetry, neurologic symptoms, or mediastinal widening. The danger is anchoring on the first clue and missing the high-risk alternative.

Feature	Pericarditis	MI	Tamponade
Pain quality	Sharp, pleuritic, positional	Pressure/heaviness, persistent	Pressure/dyspnea; may be vague
Pain relief	Sitting up and leaning forward	May improve with nitrates but not reliably	No specific positional relief
ECG	Diffuse ST elevation/PR depression	Territorial ST elevation/depression or Q waves	Low voltage or electrical alternans may occur
Key exam	Friction rub	S4, new murmur, heart failure signs	JVD, hypotension, pulsus paradoxus
Immediate danger	Effusion/tamponade	Arrhythmia, shock, mechanical rupture	Obstructive shock

High-Yield Associations for Graph-RAG

Concept	Associated nodes
HCM	Autosomal dominant, sarcomere mutation, septal hypertrophy, LVOT obstruction, Valsalva increases murmur, beta blocker, sudden death, ICD risk assessment.
Restrictive cardiomyopathy	Amyloidosis, sarcoidosis, hemochromatosis, radiation, preserved EF early, atrial enlargement, arrhythmia, congestion, distinguish from constriction.
Myocarditis	Viral prodrome, young patient, troponin elevation, chest pain, arrhythmia, heart failure, cardiac MRI, endomyocardial biopsy in selected cases, supportive care.

Acute pericarditis	Pleuritic positional pain, friction rub, diffuse ST elevation, PR depression, effusion, NSAID, colchicine, recurrent pericarditis.
Constrictive pericarditis	Rigid pericardium, impaired diastolic filling, JVD, Kussmaul sign, pericardial knock, equalized diastolic pressures, pericardiectomy.
Pericardial effusion	Echo diagnosis, low voltage, water-bottle heart if large, malignancy, uremia, hypothyroidism, pericardiocentesis if tamponade or diagnostic need.
Tamponade	Rapid fluid accumulation, hypotension, JVD, muffled heart sounds, pulsus paradoxus >10 mm Hg, electrical alternans, urgent pericardiocentesis if unstable.

Common Pitfalls

A common pitfall is treating acute pericarditis as STEMI or STEMI as pericarditis based only on ST elevation. The distribution matters. Pericarditis is usually diffuse and accompanied by PR depression, whereas STEMI localizes to a vascular territory and may have reciprocal changes. Another pitfall is ignoring tamponade because the patient lacks the complete Beck triad. Tamponade can be present even when not all classic findings are obvious, and echocardiography should be obtained quickly if the clinical picture suggests obstructive shock.

Another mistake is confusing constrictive pericarditis with cirrhosis or nephrotic syndrome. Ascites, edema, hepatomegaly, and JVD should prompt cardiac evaluation when the venous pressure is high. Similarly, restrictive cardiomyopathy and constrictive pericarditis can look nearly identical clinically; definitive differentiation may require advanced imaging or catheterization. A final pitfall is treating obstructive HCM like ordinary hypertension or stable angina. Vasodilators, dehydration, and aggressive diuresis can worsen obstruction by reducing ventricular size and increasing the gradient.

Practice Questions

Question 1

A 24-year-old athlete has exertional syncope. Exam shows a systolic murmur that becomes louder with Valsalva and standing and softer with squatting. What is the most likely diagnosis?

Answer: Hypertrophic cardiomyopathy. **Explanation:** Dynamic LV outflow obstruction worsens when preload falls. Valsalva and standing decrease LV volume and increase the murmur; squatting increases venous return and afterload, reducing the gradient.

Question 2

A 32-year-old man develops sharp pleuritic chest pain after a viral illness. Pain improves when he leans forward. ECG shows diffuse ST elevation and PR depression. What diagnostic framework applies?

Answer: Acute pericarditis. **Explanation:** At least 2 of 4 diagnostic criteria are present: typical chest pain and ECG changes. A friction rub or effusion would provide additional criteria.

Question 3

A patient with known malignancy develops dyspnea, hypotension, elevated JVP, muffled heart sounds, and an inspiratory systolic blood pressure drop of 14 mm Hg. What is the diagnosis and next step if unstable?

Answer: Cardiac tamponade; urgent pericardiocentesis if unstable. **Explanation:** Pulsus paradoxus greater than 10 mm Hg plus obstructive shock signs strongly suggests tamponade physiology.

Question 4

A patient has edema, ascites, JVD that rises with inspiration, and a pericardial knock years after chest radiation. What diagnosis should be considered?

Answer: Constrictive pericarditis. **Explanation:** Radiation can scar the pericardium. Kussmaul sign and pericardial knock reflect restricted diastolic filling.

Question 5

A patient has recurrent chest pain 1 week after an acute pericarditis diagnosis. He was treated with glucocorticoids alone. Why is this treatment choice concerning?

Answer: Glucocorticoids are associated with a higher recurrence rate and should be avoided when possible. **Explanation:** Uncomplicated idiopathic or viral pericarditis is usually treated with NSAIDs plus colchicine unless contraindications or specific causes change management.

Chapter Summary

Pericardial and myocardial diseases are best understood through physiology. HCM is a stiff hypertrophied ventricle with possible dynamic LVOT obstruction; maneuvers that reduce LV volume worsen the murmur, and therapy aims to slow the heart and improve filling. Restrictive cardiomyopathy is impaired filling due to infiltrative or fibrotic myocardium and can mimic constrictive pericarditis. Myocarditis is inflammatory myocyte injury that can mimic MI, cause troponin elevation, trigger arrhythmias, or present as acute heart failure. Acute pericarditis requires at least 2 of 4 classic criteria and is usually treated with NSAIDs plus colchicine. Constrictive pericarditis is a rigid pericardial shell causing JVD, Kussmaul sign, pericardial knock, congestion, and equalized diastolic pressures. Pericardial effusion is diagnosed best by echocardiography and may be silent until pressure rises. Tamponade is obstructive shock from impaired diastolic filling; rapid accumulation of about 200 mL can be fatal, whereas slow effusions can reach much larger volumes. Pulsus paradoxus greater than 10 mm Hg, hypotension, JVD, and echo evidence of chamber collapse are key clues, and unstable tamponade requires urgent drainage or surgery depending on cause.

Source and guideline notes

Primary coverage checklist: uploaded cardiology6.pdf, which includes cardiomyopathies, myocarditis, acute pericarditis, constrictive pericarditis, pericardial effusion, and cardiac tamponade. Additional guideline cross-checks used while drafting: 2024 AHA/ACC multisociety hypertrophic cardiomyopathy guidance; 2015 ESC pericardial disease guidance; 2024 ACC expert consensus pathway on myocarditis. This educational chapter is not a substitute for clinical guidelines or physician judgment.